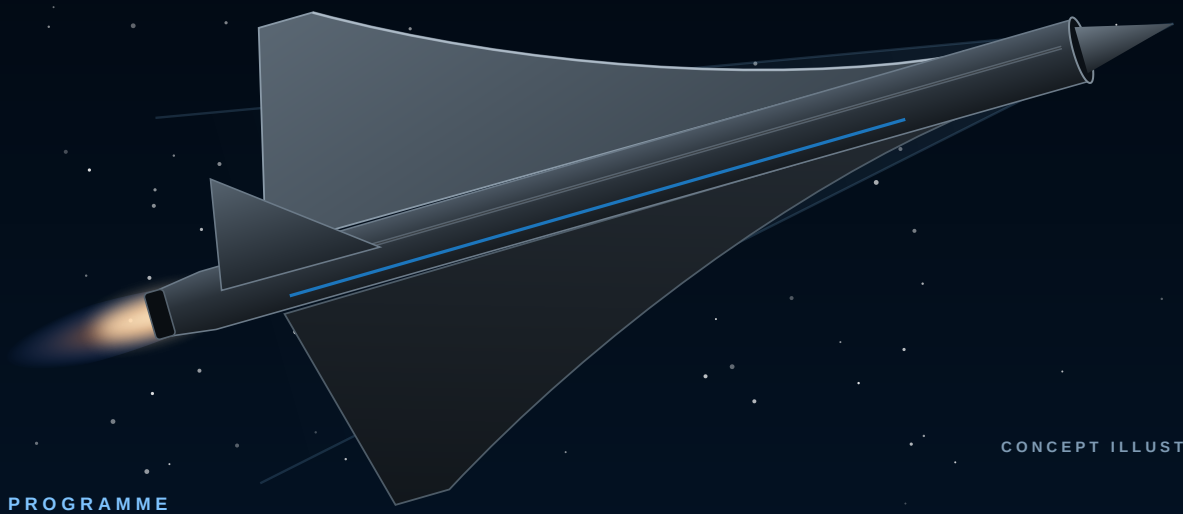
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DEFENCE AND AEROSPACE
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INDIAN TECHNOLOGY. STRONGER INDIA.



CONCEPT ILLUSTRATION

FLAGSHIP PROGRAMME

Supersonic Loitering Munition

Mach 1+ · Rotating detonation engine · Indigenous design

PROGRAMME IN DEVELOPMENT

SAG DEFENCE AND AEROSPACE · PRODUCT CATALOGUE

UNMANNED SYSTEMS & DEFENCE

DEFEND | DETER | LEAD

cto@sagdefaero.com
+91 91138 67676 +91 97408 54551

PRODUCT PORTFOLIO 2026

PRODUCT PORTFOLIO

Contents

Unmanned systems, mission modules and counter-UAS solutions, designed in India.



COUNTER-UAS
SAG NET CATCHER 03
Handheld anti-drone net catcher gun



RAZOR SERIES
RAZOR 10 04
10-inch multi-role FPV quadcopter



RAZOR SERIES
RAZOR 15 05
15-inch multi-role FPV quadcopter



RAZOR SERIES
RAZOR 18 06
18-inch heavy-lift FPV quadcopter



MISSION MODULE
RAZOR OFC 07
Fibre-optic control module, 3–30 km spools



INTELLIGENCE & SURVEILLANCE
SCOUT-X 08
Persistent ISR quadcopter, free or tethered



LOGISTICS
HANUMAN 100 09
Heavy-lift logistics UAV



MISSION MODULE
SWARM BEE 10
Plug-in swarm autonomy module



UNMANNED GROUND SYSTEMS
VAJRA 11
Autonomous tactical UGV



AIR DEFENCE
DOOM MK-1 12
Short-range surface-to-air missile



INTELLIGENCE & SURVEILLANCE
SAG VTOL-250 13
2.5 m electric fixed-wing VTOL



MARITIME · IN DEVELOPMENT
SAG TRINETRA 14
Tethered mine-neutralisation ROV



COUNTER-UAS · IN DEVELOPMENT
SAG PRAHARI 15
Body-worn drone detection system

FLAGSHIP PROGRAMME **Supersonic Loitering Munition** SEE COVER · IN DEVELOPMENT

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COUNTER-UAS & AIR DEFENCE

SAG NET CATCHER

CAPTURE | NEUTRALISE | ANALYSE

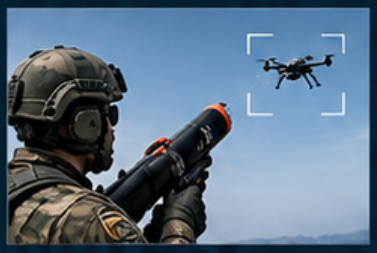
A handheld, rapid-response anti-drone net gun that fires a capture net against low-altitude aerial threats at close range.



- NON-LETHAL CAPTURE
- RAPID RESPONSE
- LIGHTWEIGHT & PORTABLE
- INTACT RECOVERY FOR INTELLIGENCE

DEFEAT
DISRUPT
DETER

HOW IT OPERATES



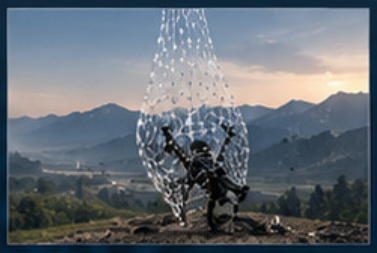
01 DETECT & AIM
Operator visually detects the hostile drone and aims the launcher.



02 LAUNCH
Press the trigger to fire a high-strength net towards the target.



03 CAPTURE
The net rapidly expands and entangles the drone, immobilising its rotors.



04 RECOVER
The captured drone falls safely to the ground for analysis and exploitation.

DRONE-TO-DRONE INTERCEPT METHOD (OPTIONAL)

The same net technology can be integrated on an interceptor drone for extended reach and higher altitude engagements.



SAG INTERCEPTOR DRONE



NET LAUNCH TOWARDS TARGET



HOSTILE DRONE CAPTURED

- EXTENDED ENGAGEMENT RANGE
- EFFECTIVE AGAINST MULTI-ROTOR UAVs
- NON-EXPLOSIVE SOLUTION
- ENABLES INTELLIGENCE RECOVERY

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
System type	Handheld net catcher gun (net launcher)
Weight	365 g
Net size	3.5 x 3.5 m
Deployment time	0.2 seconds
Effective range	25 m
Use case	Close-range anti-drone capture
Application	Static security / tactical response / force protection

IDEAL FOR

- MILITARY BASES
- CRITICAL INFRASTRUCTURE
- AIRPORTS
- PUBLIC EVENTS



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RAZOR SERIES · FPV PLATFORMS

SAG RAZOR 10

STRIKE | OBSERVE | ADAPT

A compact 10-inch FPV quadcopter for close-range strike and reconnaissance, with plug-in ISR camera modules and add-on modules for GNSS-denied and jammed environments.



- 10 KM RANGE
- 50 MIN ENDURANCE
- GNSS-DENIED READY
- JAM-RESISTANT LINK

FIND
FIX
FINISH

HOW IT OPERATES



01 LAUNCH
Operator deploys RAZOR 10 in under 5 minutes and selects the mission payload.



02 NAVIGATE
Add-on 4-element CRPA antenna module keeps GPS/ GNSS lock in jammed and spoofed environments.



03 CONNECT
Add-on frequency-hopping module keeps the control and video link alive under RF jamming.



04 EXECUTE
Operator-in-the-loop control delivers the mission — ISR, anti-personnel or anti-armour.

MISSION USE CASES



ANTI-PERSONNEL
Precision engagement of enemy positions, bunkers and trenches.



ANTI-ARMOUR
Top-attack against armoured and soft-skinned vehicles.



ISR
Day/night reconnaissance with plug-in thermal and HD camera modules.

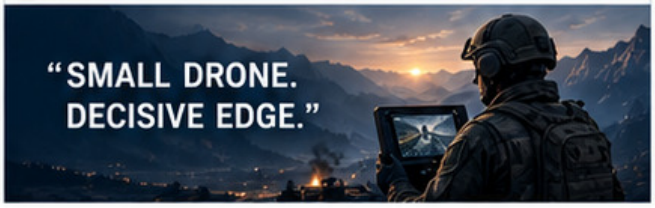
KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	FPV quadcopter — compact multirotor
Frame size	10-inch
Range	10 km
Endurance	50 min (depends on payload)
Maximum payload	1.5 kg
Mission roles	Anti-personnel, anti-armour, ISR
ISR camera modules	M1, M2, M3, M4
Add-on modules	4-element CRPA (GNSS-denied), frequency hopping (jammed RF)
Fibre-optic control	Compatible with RAZOR OFC module
Tethered operation	Compatible with SAG Tether Drone System

ADD-ON MODULES

CRPA MODULE
4-element controlled reception pattern antenna for GPS/GNSS-denied operation.

FREQUENCY HOPPING MODULE
Hops control and video link frequencies to operate through jamming.



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RAZOR SERIES · FPV PLATFORMS

SAG RAZOR 15

STRIKE | INTERCEPT | SUSTAIN

A 15-inch multi-role FPV quadcopter for strike, reconnaissance, counter-drone net capture and cargo delivery, with add-on modules for GNSS-denied and jammed environments.



- 15 KM RANGE
- 30+ MIN ENDURANCE
- GNSS-DENIED READY
- JAM-RESISTANT LINK

FIND
FIX
FINISH

HOW IT OPERATES



01 CONFIGURE
Operator fits the mission payload: warhead, ISR camera, net launcher or cargo pod.



02 NAVIGATE
Add-on 4-element CRPA antenna module keeps GPS/GNSS lock in jammed and spoofed environments.



03 CONNECT
Add-on frequency-hopping module keeps the control and video link alive under RF jamming.



04 EXECUTE
Operator-in-the-loop control delivers the mission across a 15 km range.

MISSION USE CASES



ANTI-PERSONNEL
Precision engagement of enemy positions and bunkers.



ANTI-ARMOUR
Top-attack against armoured vehicles.



ISR
Stabilised day/night observation up to 4K.



C-UAS NET LAUNCHER
Carries a net launcher to capture hostile drones in flight.



CARGO
Rapid resupply of forward posts.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	FPV quadcopter — medium multirotor
Frame size	15-inch
Range	15 km
Endurance	30+ min (depends on payload)
Maximum payload	3 kg
Mission roles	Anti-personnel, anti-armour, ISR, C-UAS net launcher, cargo
ISR camera modules	M4, M5, M6
Add-on modules	4-element CRPA (GNSS-denied), frequency hopping (jammed RF)
Fibre-optic control	Compatible with RAZOR OFC module
Tethered operation	Compatible with SAG Tether Drone System

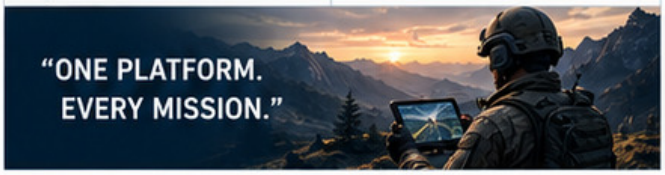
ADD-ON MODULES



CRPA MODULE
4-element controlled reception pattern antenna for GPS/GNSS-denied operation.



FREQUENCY HOPPING MODULE
Hops control and video link frequencies to operate through jamming.



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RAZOR SERIES · HEAVY-LIFT FPV

SAG RAZOR 18

LIFT | STRIKE | DESIGNATE

The heavy-lift member of the RAZOR series: an 18-inch FPV quadcopter carrying up to 8 kg for strike, full-sensor ISR, counter-drone net capture and cargo, with add-on modules for GNSS-denied and jammed environments.



- 20 KM RANGE
- 8 KG PAYLOAD
- GNSS-DENIED READY
- JAM-RESISTANT LINK

FIND
FIX
FINISH

HOW IT OPERATES



01 CONFIGURE
Operator fits the mission payload up to 8 kg: warhead, ISR gimbal with laser range finder, net launcher or cargo pod.



02 NAVIGATE
Add-on 4-element CRPA antenna module keeps GPS/GNSS lock in jammed and spoofed environments.



03 CONNECT
Add-on frequency-hopping module keeps the control and video link alive under RF jamming.



04 EXECUTE
Operator-in-the-loop control delivers the mission across a 20 km range.

MISSION USE CASES



ANTI-PERSONNEL
Precision engagement of enemy positions and bunkers.



ANTI-ARMOUR
Heavy top-attack payload against armoured vehicles.



ISR
Full-sensor ISR and target designation with laser range finder.



C-UAS NET LAUNCHER
Carries a net launcher to capture hostile drones in flight.



CARGO
Heavy resupply of high-altitude and forward posts.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	FPV quadcopter — heavy-lift multirotor
Frame size	18-inch
Range	20 km
Endurance	30+ min (depends on payload)
Maximum payload	8 kg
Mission roles	Anti-personnel, anti-armour, ISR, C-UAS net launcher, cargo
ISR camera modules	M5, M6, M7, M8
Add-on modules	4-element CRPA (GNSS-denied), frequency hopping (jammed RF)
Fibre-optic control	Compatible with RAZOR OFC module
Tethered operation	Compatible with SAG Tether Drone System

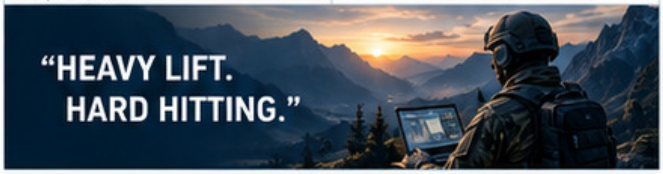
ADD-ON MODULES



CRPA MODULE
4-element controlled reception pattern antenna for GPS/GNSS-denied operation.



FREQUENCY HOPPING MODULE
Hops control and video link frequencies to operate through jamming.



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RAZOR SERIES · MISSION MODULE

SAG RAZOR OFC

RF-SILENT | JAM-RESISTANT | MISSION-READY

A fibre-optic control module family for RAZOR 10, RAZOR 15 and RAZOR 18. It replaces the radio link with a fibre-optic spool for RF-silent, jam-resistant real-time control and video in contested and heavily jammed environments.

RF-SILENT LINK

JAM-RESISTANT

LOW-LATENCY VIDEO

3–30 KM SPOOLS

CONNECT
CONTROL
CONQUER

CHOOSE YOUR RANGE



One module family. Spool length matched to mission range and platform.

HOW IT OPERATES



01 SELECT
Choose the spool length to match the mission range.



02 MOUNT
Fit the OFC module to RAZOR 10, RAZOR 15 or RAZOR 18.



03 FLY OVER FIBRE
Control and video travel through the fibre. No radio emission from the control



04 OPERATE UNDER JAMMING
The link stays clear where radio-controlled drones lose control.

MISSION USE CASES



COUNTER-EW MISSIONS
Operate in heavily jammed front-line zones.



URBAN RECONNAISSANCE
Fly between buildings without an RF signature.



TUNNEL & UNDERGROUND
Keep control and video where radio signals cannot reach.



BORDER & PERIMETER
Silent surveillance of borders and critical sites.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Module type	Fibre-optic control and video module (spool)
Compatible platforms	RAZOR 10, RAZOR 15, RAZOR 18
Fibre length options	3 km, 5 km, 7 km, 10 km, 15 km, 20 km, 25 km, 30 km
Control and video link	RF-silent real-time control and video over fibre
Guidance	Operator-in-the-loop control over fibre
EW resilience	Unaffected by RF jamming and spoofing on the control/video bto link
Integration role	Mission module for contested-environment operations

RAZOR PLATFORM COMPATIBILITY



RAZOR 10
Lightweight. Agile. Versatile.



RAZOR 15
More payload. Greater reach.



RAZOR 18
Heavy lift. Maximum capability.

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INTELLIGENCE & SURVEILLANCE

SAG SCOUT-X

WATCH | TRACK | PERSIST

An ISR quadcopter combining free flight and continuous tethered operation for persistent monitoring and mission-critical border security.

90 MIN FREE FLIGHT

24+ HR TETHERED

20 KM RANGE

GNSS-DENIED NAVIGATION

OBSERVE
DETECT
DENY

HOW IT OPERATES



01 DEPLOY
VTOL launch; ready in under 5 minutes.



02 PATROL
Free flight for up to 90 minutes, or tethered for 24+ hours with the SAG Tether Drone System.



03 OBSERVE
Full HD day camera and 640 x 512 thermal EO/IR, day and night.



04 SHARE
Encrypted AES-256 full-motion video and telemetry to the ground station.

MISSION USE CASES



BORDER SURVEILLANCE
Persistent watch over long border stretches.



COUNTER-INFILTRATION
Thermal detection of movement at night.



ROUTE RECONNAISSANCE
Clear routes ahead of patrols and convoys.



POST OVERWATCH
Tethered 24+ hour overhead cover for forward posts.

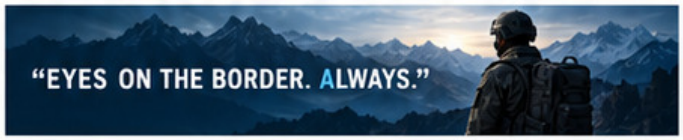
KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	ISR quadcopter
Range	20 km
Endurance	90 min free flight; 24+ hr tethered (extendable per mission)
Maximum take-off weight	8 kg
Payload	Up to 1.5-2 kg modular ISR payload
Maximum speed	60 km/h
Operational altitude	Up to 5,000 m AMSL
Day / night camera	1920 x 1080p (2K/4K options) / 640 x 512p thermal EO/IR
Navigation	GNSS-denied navigation through jamming and spoofing
Data link	Encrypted RF, AES-256; mesh-capable multi-asset coordination
Protection / standards	IP55; MIL-STD-461E; MIL-STD-810G

TETHERED MODE



Connect the SAG Tether Drone System for continuous 24+ hour operation.



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LOGISTICS & COORDINATED UAVS

SAG HANUMAN 100

LIFT | CARRY | DELIVER

A heavy-lift aerial-delivery platform for defence logistics and humanitarian missions, including supply to locations with limited ground access.



- 80 KG PAYLOAD
- 10 KM RANGE
- RTK PRECISION LANDING
- 5,000 M ALTITUDE

SUPPLY
SUSTAIN
SUPPORT

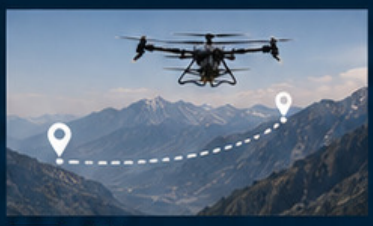
HOW IT OPERATES



01 LOAD
Load up to 80 kg of supplies into the cargo bay.



02 LIFT OFF
VTOL launch; ready in under 5 minutes.



03 FLY
GNSS waypoint navigation with obstacle avoidance; live video and payload status.



04 DELIVER
RTK precision landing at the destination post.

MISSION USE CASES



HIGH-ALTITUDE POST RESUPPLY
Food, fuel and ammunition to snow-bound posts.



FORWARD OUTPOST LOGISTICS
Replaces risky mule and porter runs.



MEDICAL SUPPLY
Urgent medicines and blood to remote units.



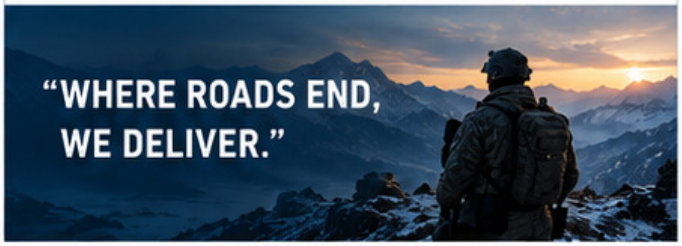
HUMANITARIAN RELIEF
Aid delivery where roads are cut off.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	Heavy-lift logistics UAV
Range	10 km
Endurance	40-50 min unloaded; 25-35 min at maximum payload
Maximum take-off weight	200 kg
Payload	Up to 80 kg
Speed	60 km/h
Operational altitude	Up to 5,000 m AMSL
Navigation	GNSS waypoint, RTK precision landing, obstacle avoidance; EW compliant
Data link	Encrypted RF, AES-256; telemetry, payload status and live video
Launch	VTOL; under 5 minutes to set up
Protection / standards	IP55; MIL-STD-461E; MIL-STD-810G

IDEAL FOR

- MOUNTAIN POSTS
- ISLAND & COASTAL UNITS
- DISASTER ZONES
- FORWARD BASES



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MISSION MODULE · SWARM AUTONOMY

SAG SWARM BEE

CONNECT | COORDINATE | CONVERGE

A plug-in swarm module that turns any SAG drone into a node of an AI-enabled swarm. Drones share targets, split the area, avoid each other and strike from every direction at once.

- PLUG-IN MODULE
- ALL SAG PLATFORMS
- SELF-HEALING MESH
- 360° STRIKE

SWARM
SURROUND
STRIKE

HOW IT OPERATES



01 FIT
Fit the SWARM BEE module to any SAG drone.



02 LINK
Drones join an encrypted, self-healing mesh and share position and targets.



03 ASSIGN
AI allocates tasks and areas dynamically across the swarm.



04 CONVERGE
Drones approach the target together from multiple directions.

STRIKE FROM EVERY DIRECTION



Simultaneous multi-axis attack overwhelms a target's defences and leaves no safe side.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Module type	Plug-in swarm autonomy module
Compatible platforms	RAZOR 10, RAZOR 15, RAZOR 18, SCOUT-X, SAG VTOL-250
Swarm intelligence	Centralised, distributed or hybrid
Autonomy level	Assisted / semi-autonomous / fully autonomous
Formation modes	V-shape, line, circular and adaptive clustering
Collision avoidance	Inter-drone and obstacle avoidance
Task allocation	Dynamic mission assignment
Strike geometry	Simultaneous multi-directional: front, back, left, right, top
Communication / data link	Encrypted, self-healing mesh radio; peer-to-peer relay across the swarm
Launch and deployment	Sequential or simultaneous deployment

SAG PLATFORM COMPATIBILITY

- RAZOR 10
- RAZOR 15
- RAZOR 18
- SCOUT-X
- SAG VTOL-250



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UNMANNED GROUND SYSTEMS

SAG VAJRA

CARRY | CLEAR | COMBAT

An autonomous tactical ground platform for multi-mission support, evacuation and hazardous-area operations across varied terrain.



500 KG PAYLOAD | 100 KM RANGE | 3-4 HR ENDURANCE | AI NAVIGATION

MOVE
PROTECT
PREVAIL

HOW IT OPERATES



01 PLAN
Operator sets the mission; AI plans and optimises the route.



02 NAVIGATE
All-terrain chassis drives autonomously with machine-learning threat recognition.



03 EXECUTE
Carries out the fitted mission: evacuation, EOD, missile launch or armed overwatch.



04 REPORT
Encrypted real-time telemetry and extended-range operator command.

MISSION USE CASES



CASUALTY EVACUATION
Autonomous evacuation with tactical route planning.



EXPLOSIVE ORDNANCE
Mine detection and controlled disposal by remote operation.



C-UAS LAUNCH PLATFORM
Mobile short-range counter-drone missile launcher.



LOGISTICS
500 kg of supplies to the front line.



BORDER SECURITY
Autonomous patrol of borders and perimeters.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	AI-enabled autonomous tactical UGV
Operational range	100 km
Endurance	3-4 hours
Payload capacity	500 kg
Navigation	AI-based navigation, autonomous path planning, adaptive route optimisation
AI intelligence	ML threat recognition, terrain analysis, mission-adaptive behaviour
Mobility	High-mobility all-terrain chassis, adaptive suspension
Sensor suite	Integrated multisensor array and threat detection
Mission modules	Evacuation; EOD; C-UAS missile launch; rifle mount with autonomous target acquisition
Communication	Encrypted tactical comms, real-time telemetry, extended-range command
Power	Autonomous battery management and endurance optimisation

MISSION MODULES

EVACUATION

EOD / DEMINING

MISSILE LAUNCHER

WEAPON STATION



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COUNTER-UAS & AIR DEFENCE

SAG DOOM MK-1

DETECT | LAUNCH | DESTROY

A short-range air-defence missile system integrating a vehicle-mounted multi-canister launcher and rapid-response engagement capability.



2-5 KM RANGE

<7 S REACTION

<2 M CEP

5,000 M CEILING

DEFEND
DENY
DESTROY

HOW IT OPERATES



01 DETECT
A hostile UAV is detected and tracked.



02 LAUNCH
Missile leaves the truck-mounted canister in under 7 seconds from detection.



03 GUIDE
Dual-pulse solid rocket motor with fin control steers to the target.



04 INTERCEPT
Proximity fuze detonates the pre-fragmented HE warhead at the target.

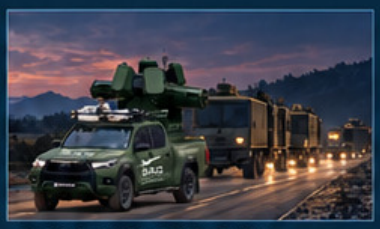
MISSION USE CASES



AIR BASE DEFENCE
Point defence of airfields against hostile drones.



FORWARD POST PROTECTION
Rapid response for high-altitude posts.



CONVOY PROTECTION
Mobile launcher moves with the convoy.

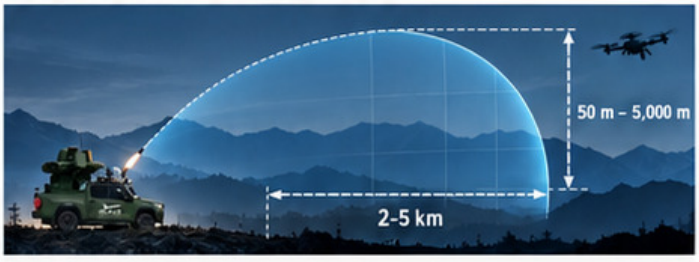


CRITICAL INFRASTRUCTURE
Shield power, fuel and command sites.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	Short-range surface-to-air missile
Mission role	Air defence against UAVs
Engagement range	2-5 km extended short-range engagement envelope
Engagement altitude	50 m to 5,000 m
Accuracy	CEP <2 m at maximum range
Propulsion system	Dual-pulse solid rocket motor with fin control
Warhead	Pre-fragmented HE, 0.5-1.5 kg, with proximity fuze
Launch configuration	Truck-mounted multi-canister launcher
Reaction time	<7 seconds from target detection to launch

ENGAGEMENT ENVELOPE





SAG
DEFENCE AND AEROSPACE
नवोन्मेष तन्त्रज्ञाना भ्यां राष्ट्रसेवा

INDIAN TECHNOLOGY. STRONGER INDIA.

INTELLIGENCE & SURVEILLANCE

SAG VTOL-250

RISE | CRUISE | WATCH

A 2.5 m carbon-fibre electric VTOL for long-endurance standoff ISR and wide-area surveillance. It takes off and lands vertically with no runway or launcher, then cruises on the wing for up to 4 hours.

- **4 HR ENDURANCE**
- **2.5 M WINGSPAN**
- **NO RUNWAY**
- **4,800 M ALTITUDE**

SEE FAR BEYOND

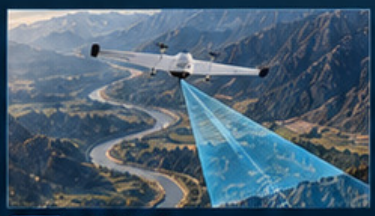
HOW IT OPERATES



01 TAKE OFF
Vertical take-off from any clearing; no runway or launcher.



02 TRANSITION
Converts to efficient fixed-wing cruise.



03 SURVEIL
Wide-area ISR with M6 4K zoom, M7 hybrid zoom or M8 dual-sensor with LRF.



04 RECOVER
Returns and lands vertically at the launch point.

MISSION USE CASES



WIDE-AREA BORDER SURVEILLANCE
Hours on station over long border stretches.



STANDOFF ISR
Observe from a distance, outside threat range.



COASTAL PATROL
Long-endurance watch over coastline and creeks.



CONVOY ROUTE OVERWATCH
Scan the route ahead of moving columns.

KEY SPECIFICATIONS

PARAMETER	SPECIFICATION
Platform type	Electric fixed-wing VTOL
Wingspan / length	2,500 mm / 1,260 mm
Airframe	Carbon fibre
Maximum take-off weight	15 kg (0-3,000 m)
Recommended payload	1.2 kg (excluding battery)
Endurance	4 h no payload; 3.5 h with 1.2 kg payload
Speed	Max cruise 26 m/s (≈94 km/h); stall 15.5 m/s (≈56 km/h)
Maximum flight altitude	4,800 m
Wind resistance	Fixed-wing 10.8-13.8 m/s; VTOL 5.5-7.9 m/s
Operating temperature	-20 to +45 °C
ISR camera modules	M6, M7, M8
Transport case	1,300 x 530 x 470 mm

ISR PAYLOAD OPTIONS

M6



4K / 8 MP, 30× digital zoom

M7



2K / 4 MP, 30× hybrid (10× optical)

M8



2K + 640×512 thermal, LRF 905 nm, 5 m – 1 km



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PROGRAMME IN DEVELOPMENT

MARITIME SYSTEMS • MINE COUNTERMEASSURES

SAG TRINETRA

LOCATE | NEUTRALISE | REPEAT

An indigenous, reusable, tethered underwater vehicle that neutralises multiple sea mines in a single sortie, without surfacing or reloading between targets. Platform-agnostic: it launches from any compatible vessel with a standard winch or davit.



MULTI-SHOT SORTIE | TETHER-POWERED | 300 M DEPTH | HUMAN-IN-THE-LOOP

CLEAR
SECURE
PROTECT

HOW IT OPERATES



01 DEPLOY
Launched from the host vessel by winch or davit to a reported mine contact.



02 REACQUIRE
Onboard sonar and camera relocate the contact; the operator confirms the target.



03 PLACE & RETREAT
The ROV places a neutralisation charge and retreats to a safe standoff distance.



04 NEXT TARGET
It proceeds straight to the next confirmed contact — up to 3 per sortie — with no resurfacing.

INTEGRATED MINE COUNTERMEASURES



Operated from SAG MATSYA USV or any compatible naval or coast guard vessel.

DESIGN TARGETS

PARAMETER	TARGET
System type	Tethered, reusable mine-neutralisation ROV
Operational depth	300 m
Vehicle length	Approx. 2.5–3 m
Forward speed	Up to 5–6 knots
Charges per sortie	Up to 3 modular charge modules
Mission endurance	Effectively unlimited (tether-powered)
Sensors	High-frequency sonar and camera reacquisition suite
Control	Remote operator station; human confirmation at every engagement
Launch and recovery	Platform-agnostic; standard shipboard/USV winch or davit
Indigenous content	70%+ at prototype, 85%+ at final delivery

IDEAL FOR

HARBOUR & PORT CLEARANCE | SEA LANE & CHANNEL CLEARANCE | MCM VESSELS & USVs | COASTAL SECURITY



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PROGRAMME IN DEVELOPMENT

COUNTER-UAS · DETECTION

SAG PRAHARI

DETECT | TRIANGULATE | ALERT

A body-worn, dual-modality drone detection system for the dismounted squad. Wideband SDR catches radio-controlled drones; a passive acoustic array catches RF-silent, fibre-optic drones; a squad mesh fuses bearings into one located threat.

30 MHz–6 GHz SDR

ACOUSTIC OFC DETECTION

SQUAD TRIANGULATION

ZERO-EMISSION PASSIVE

SENSE
SHARE
SURVIVE

HOW IT OPERATES



01 DETECT
Each unit sweeps 30 MHz–6 GHz by SDR and listens for rotor harmonics acoustically.



02 SHARE
Type, bearing, GPS position and time are shared across the squad's LoRa mesh.



03 TRIANGULATE
Bearings from multiple soldiers are fused into one computed ground position.



04 ALERT
Every soldier gets the bearing on the chest display with an audible and haptic alert.

CLOSING THE DETECTION GAP

RF-ONLY DETECTORS	SAG PRAHARI
Fixed narrow-band scan	Continuous 30 MHz – 6 GHz
Blind to fibre-optic drones	Acoustic catches RF-silent drones
Single-point bearing	Squad-level triangulated fix

SYSTEM COMPONENTS



BACKPACK SENSOR POD



CHEST-MOUNTED DISPLAY

DESIGN TARGETS

PARAMETER	TARGET
System type	Body-worn, squad-networked C-UAS detection
RF / SDR coverage	30 MHz – 6 GHz (continuous)
RF detection range	> 1 km
Acoustic detection range	150–300 m typical (up to 500 m in low-noise conditions)
Acoustic band	70 Hz – 2000 Hz rotor harmonics
Squad networking	LoRa mesh, 8–10 units at 50–100 m spacing
Multi-target tracking	≥ 10 simultaneous contacts (RF channel)
Total system weight	1.2 – 1.8 kg (distributed)
Operating mode	Fully passive, receive-only (EMCON capable)
Indigenous content	70%+ at prototype, 85%+ at final delivery

IDEAL FOR

COUNTER-TERROR OPERATIONS

FORWARD POSTS & PICKETS

PATROLS & CONVOYS

MOUNTAIN & URBAN TERRAIN

